Application of Queue Theory to Waiting Line Problems In Emergency Department of The Lagos State General Hospital, Odan, Lagos Island, Lagos.

A.B. Sogunro and B. Abiola
amudasogunro@yahoo.co.uk and abankole2006@yahoo.com

Department of Actuarial Science and Insurance,
Faculty Business Administration,
University of Lagos,
Akoka, Yaba, Lagos.Nigeria

Abstract

Much research effort has been devoted to improving both medical and management practices of health care services. The healthcare processes which vary in complexity and scope consist of a set of activities and procedures which the patient must undergo to receive the required treatment. Unfortunately, even with improved technology, patients still experience delays and waiting, which, sometimes, lead to dissatisfaction and consequently violence among waiting patients. This study attempts to find a means of reducing the length of the waiting line of patients in the Odan General Hospital emergency department in process of consulting doctors using queuing analysis to improve the current level of performance and identify other opportunities for the same purpose. Using observed data gathered for two weeks during the morning shift (from 8.00 am to 4.00pm), the result of the sensitivity analysis of the study evaluated by using number of servers (doctors) against the four queuing performance metrics showed that a patient would spend less time on the queue and system if the number of servers can be increased to two and above especially during the peak period of consultation. Therefore, full integration of queue theory applications to the routine structures and practices of the emergency department of the hospital will allow waiting time targets to be attained.

Keywords: Sensitivity analysis, Queuing, Emergency, System, Server, Performance metrics and Healthcare.

Introduction

The ability of the healthcare system to serve patients quickly, reliably and efficiently as they move through stages of care is an important issue that needs urgent attention.

Healthcare centre represents an area where queuing models can be effectively used as tools in evaluating patients' access, care policies and efficiencies (McLeod,et-al 2009). The healthcare services provided by an organization that takes care of the ill and injured people can be viewed as queuing systems in which patients arrive, wait for service receive, service and then depart. The process varies in complexity and scope, and consists of a set of activities and procedures (both medical and non-medical) that the patient must undergo to receive the needed treatment. The resources (or servers) in these queuing systems are trained personnel and specialized equipment that these activities and procedures require (Fomundam and Herrmann, 2007).

Waiting time has been mentioned by many researchers as the most important cause of dissatisfaction, delays and increased violence of patients attending emergency departments (ED)(Trout et al 2000). The amount of time patients must spend waiting can significantly influence their satisfaction (Davis and Volkmann, 1990). Patient satisfaction is not only affected by waiting time but also by patient expectations or attribution for the causes of the waiting (Taylor, 1994).

Queuing, one of the challenges of most hospitals, has become a symbol of inefficiency in publicly funded hospitals in the world, Nigeria inclusive (Belson, 1988). Managing the length of the line may be owing to a lack of commitment on part of hospital staff, overloading of available doctors and unavailability of resources (Obamiro, 2010). These problems put doctors under stress and pressure and lead to dismissing of patients without adequate treatment which lead to patients' dissatisfaction (Babes and Sharma, 1991).

After World War II(1939-1945) queuing theory became an area of research interest. Although, much has been researched on queuing theory and its importance, its application to real life situations, it was only recently that healthcare experts discovered the benefits of applying queuing theory techniques. Not much work has been done especially in area of emergency department of the healthcare.

Queuing theory, a living branch of science which is extensively practiced or utilized in operations management, was first analyzed by a Danish engineer A.K. Erlang in 1909 to model the number of telephone calls arriving at an exchange by a Poisson process and used by Felix Pollaczek to solve the M/G/1 queue in 1930 (Kingman, 2009). Erlang works inspired mathematicians and others related discipline to deal with queuing problems using probabilistic methods. Queuing theory became a field of applied probability and many of its results have been used in operations research, telecommunication, traffic problems and field of sciences.

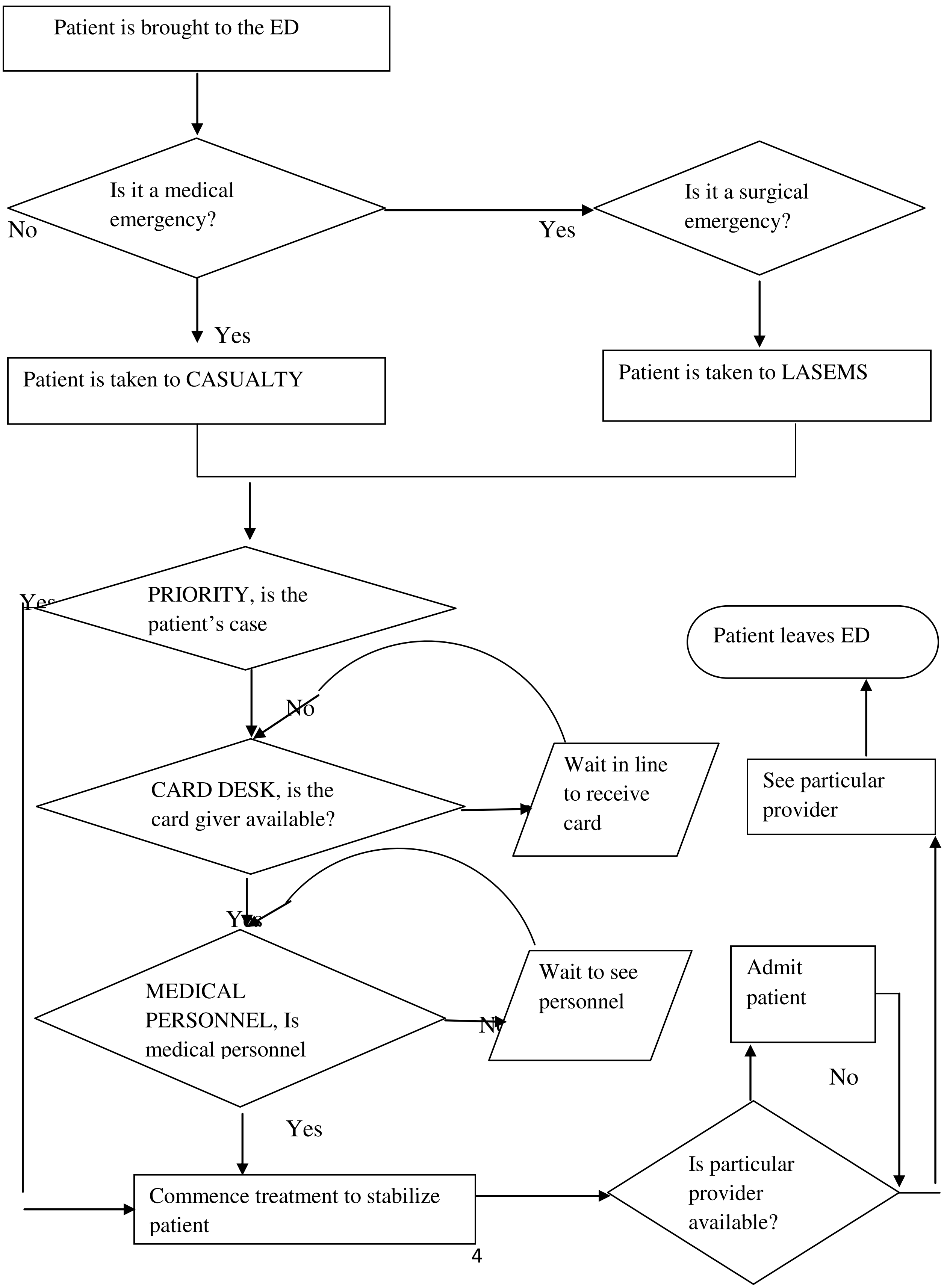
Queuing theory can be used to solve many problems in the emergency section of the healthcare system, such as: patient long waiting time, patient treatment procedures, scheduling appointment, resources utilization and many others, and has been used extensively by the service industries (Nosek and Wilson, 2001). Its application in health care is an attempt to minimize the cost of providing healthcare services through minimization of efficiencies and delay in the system (Singh 2006).

In most Nigerian public hospitals, emergence department of a hospital is one of the busiest and most congested as compared to other departments of the hospital. Patients have to wait for a long time to receive treatments, both on queue and during service which brought about decrease in patient satisfaction, decrease in quality of care, economic cost of the organization and increase in cost owing to delay in care or the value of patients time. These situations can be improved with advancement in technology and continuous adjustments of healthcare systems using queuing analysis which can help in reducing the bottlenecks and maintain proportion between resources and service delays.

Viewing the General hospital, emergency department, Odan as a closed unit, patients arrive and leave after receiving treatment. Arriving patients are attended to on a first come basis or treated according to priority. While medical emergencies are sent to the casualty division, surgical emergencies are sent to the Lagos State Emergency Management Agency (LASEMA) division. Patients are treated and admitted if need be and from there transferred to their respective section as shown in figure I.

This study was carried out within the scope of the emergency department of the General hospital, Odan where we could record an average of six hundred out-patients per day using data gathered for two weeks of observation in morning shift (from 8.00 am to 4.00pm).

Figure I **Patient flow through the emergency department**



Methodology

Queuing theory is concerned with the statistical description of the behavior of queues with finding the probability distribution of the number in the queue from which the mean and variance of queue length and the probability distribution of waiting time for a customer or the distribution of a server's busy periods can be found (Sharma, 2010).

There are three types of queuing model; probabilistic, mixed and deterministic queuing models, and the usage of which depend on the features that are significant in specific situations.

Probabilistic queuing models are further divided into Erlang, General Erlang, Model iii, iv, v… ix, each of which is used for different functions depending on the features that are significant to specific situation. This significant feature has to be incorporated into the model.

Queuing analysis which is used by managers to make decision is known to be probabilistic analysis. Sometimes one needs to change the system to alter its performance after calculating the operating characteristics of a waiting line.

The operating characteristics of queuing systems are determined largely by two statistical properties; probability distribution of inter-arrival times (exponential distribution) and the probability distribution of service times (Poisson process). In real queuing systems, these characteristics can take on any form, (except negative characteristics). However to formulate a queuing theory model as a representation of the real system, it is necessary to specify the assumed form of each of these distributions.

The operating characteristics of a waiting line system are calculated using the following formulas:

λ = mean arrival rate of customers (average number of customers arriving per unit of time)

μ = mean service rate (average number of customers that can be served per unit of time)

$\rho = \frac{\lambda}{\mu}$ = the average utilization of the system (Utilization factor or Traffic Intensity)

$L = \frac{\lambda}{\lambda - \mu}$ = the average number of customers in the service system

$L_Q = \rho L$ = the average number of customers waiting in line

$W = \frac{1}{\mu - \lambda}$ = the average time spent waiting in the system including serviceType equation here.

$W_Q = \rho W$ = the average time spent waiting in line

$P_n = (1-\rho^n)\rho^n$ = the probability that n customers are in the service system at a given time

Where $\mu > \lambda$

A multiple-server waiting line system assumes s identical servers, the service time distribution for each server is exponential and the mean service time is 1/μ. Using these, the operating characteristics are defined as follows.

s = number of servers in the system

$\rho = \frac{\lambda}{s\mu}$ = average utilization of the system

P_0 = the probability that no customers are in the system

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{(\lambda/\mu)}{n!} + \frac{\left(\frac{\lambda}{\mu}\right)}{s!}\left(\frac{1}{1-\rho}\right)\right]^{-1}$$

L_q= average number of customers waiting in line

$$L_q = \frac{P_0(\lambda/\mu)^s\rho}{s!\,(1-\rho)^2}$$

W_q= the average time spent in the line

$$W_q = \frac{L_q}{\lambda}$$

W= the average time spent in the system, including service

$$W = W_q + \frac{1}{\lambda}$$

L= the average number of customers in the system

$$L = \boldsymbol{\lambda} W$$

P_n = probability that n customers are in the system at a given time

$$P_n = \begin{cases} \frac{(\lambda/\mu)^n}{n!}P_0 & for\ n \leq s \\ \frac{(\lambda/\mu)^n}{s!\,s^{n-s}}P_0 & for\ n > s \end{cases}$$

The total service rate must be greater than the arrival rate i.e. sμ > **λ**. If sμ ≤ **λ**, the waiting line would eventually grow infinitely large.

Two important probability distributions called Markov distributions (Poisson and Exponential Distributions) are often used in queuing models due to their significant features and ability for random processes..

Poisson process

The Poisson distribution is a good approximation for many processes that are considered random (number of events in a unit of time) and it assumes an infinite number of possible arrivals, with each having a small probability of occurrence. Arrivals are independent and the number of arrivals in one period of time does not affect the number in the next. The arrival pattern of patients at a queuing system varies between one system and another, but one pattern of common occurrence in practice which turns out to be relatively easy to deal with mathematically, is that of completely random arrivals. If the arrivals are completely random, then the probability distribution of number of arrivals in a fixed-time interval follows a Poisson distribution. If the patient's arrival process follows a Poisson probability distribution, the corresponding patients' inter-arrival process will follow the exponential probability distribution.

Assumptions:

To derive the arrival distribution in queues, the following assumptions are made:

.That there are n units in the system at time t, and the probability that exactly one arrival will occur during small time interval Δt be given by $\lambda\Delta t+o(\Delta t)$, where λ is the arrival rate independent of t and $o(\Delta t)$ includes the terms of higher order of Δt.

.The time Δt is so small that the probability of more than one arrival in time Δt is $o(\Delta t)^2$, i.e. almost zero.

.The number of arrivals in non-overlapping intervals is statistically independent, i.e., the process has independent increments

.Thus the probability of n arrivals in time t follows the Poisson law given by the equation;

$P_n(\mathrm{t}) = (\lambda t)$

Exponential probability

Exponential distribution is a complementary distribution to the Poisson that has as the random variable the time between events. It will effectively model the random arrival of patients during a given period of time **t** at the emergency room by considering that the probability of **x** patients in a time period of length with a queue discipline first come first served adhere to the following assumptions;

- The input process is assumed to be finite, i.e. arrival pattern is completely random
- Probability of more than one customer arriving during a short period of time t must be essentially zero
- The number of customer arrivals in two different are independent
- Queue discipline is first come first serve (FCFS)
- Unit time is in minute
- Allowable maximum length of the waiting line is infinite in size
  Mohammad Shyfur (2013).

Most models conventionally, are usually described using David G. Kendall's notation (a/b/c) (Sanjay, 2002) In 1966, A.M. Lee added the symbols d and e to the notation, while Hamdy A. Taha added the last element, symbol f in 1968 to make the notation become (a/b/c/d/e/f) (Hamdy, 2007). Consider the single server, finite capacity queue with Poisson arrivals and generally distributed service times. The M/M/1/K system may be analysed using an imbedded Markov Chain approach. A queue of this type may be a better representation of a real-life system and can provide equilibrium results. This is because the infinite number of buffers implied by the M/M/1/∞ model would be difficult to satisfy in a real system, except probably as an approximation.

To facilitate proper data analysis for the (M/M/2/∞/FCFS) system under study, (That is, two servers queue with Poisson arrivals, exponentially distributed service times and infinite number of waiting positions with a queue discipline first come first served), It is essential to buildup models based on the following assumptions;

- Patient arrival pattern is a Poisson process (since arrival pattern is a discrete event).
- Servicing process follows exponential probability (Because service rate is a continuous distribution).

Method

A week was spent in each of the two divisions of the (ED) emergency departments (casualty and emergency service) of the hospital for direct observation of the field and collection of the needed primary data. A stop watch and a wall clock for the accuracy of time with some trained personnel's needed for proper collection and recording of relevant information such as number of patients, the arrival times of patients, waiting time and service times were used. The observation was made for ten working days (morning shift between the hours of 8a.m and 4p.m).The observed information was recorded and then transferred to an excel sheet for further calculations of average waiting time, average service time and the utilization factor.

The information gathered for the ten days of the field observation between the hours of 8am to 4pm was analyzed using queuing analytical techniques. Using the observed data with two servers, inter-arrival time and service time of each patient for each day was computed. With the standard queuing function, the average inter-arrival time, average service time, standard deviation of both the inter-arrival time and service time was then calculated on excel format.

The result of the average inter-arrival time, average service time and standard deviation of both inter-arrival time and service time were then used to compute the arrival rate, service rate and the queuing performance metrics, that is, average number of patients in the system (L), Average time spent by patient both on queue and in service (W), Average time spent by the patient (Wq) and average number of patients on queue waiting for service (Lq) for each day throughout the observation period.

Results and Discussion

After obtaining the queuing performance metrics (L, Lq, W, Wq) using two servers (two doctors attending to the patients), a sensitivity analysis was then carried out on the number of servers against the four queuing performance metrics for each day of the observation to know the effect of change of servers on the four queuing performance metric. The generated sensitivity analysis results for each value of the queuing performance metrics is given in the table I, II, III and IV below.

Table I, shows the results of the sensitivity analysis of numbers of servers against average numbers of patients both on queue and in service (L) for each day. For example, column L_1 row 1 (two servers) indicate an average number of patient both on queue and in service (i.e. 68.16521) for the first day. Therefore, In general column L_1, L_2, L_3, L_4, L_5, L_6 , L_8 , L_9, L_{10} of row 1 (2 servers) show that 68.165, 26.011, 54.99, 62.26, 62.04, 25.743, 106.924, 22.23 and 2.998, respectively, represent average number of patients both on the queue and in service. The fluctuation in the number of sever against the average number of patients for the ten days of the observation in the (ED)emergency unit of the hospital is demonstrated in figure II below where average number of patients both on queue and in service (Ls) for each day are represented with different colour lines of the graph.

Table I: Results of the sensitivity analyses of number of servers against average number of patients

server	L_1	L_2	L_3	L_4	L_5	L_6	L_8	L_9	L_{10}
2	68.165	26.012	54.994	62.621	62.621	25.743	106.924	22.233	29.998
3	54.808	24.096	45.900	52.087	52.087	24.221	82.878	21.639	28.095
4	48.130	23.138	41.353	47.000	47.000	23.460	70.855	21.326	27.143
5	44.123	22.564	38.625	43.948	43.948	23.003	63.641	21.145	26.572
6	41.451	22.181	36.806	41.914	41.914	22.699	58.832	21.024	26.192
7	39543	21.907	35.507	40.460	40.460	22.481	55.400	20.938	25.920
8	38.112	21.702	34.533	39.370	39.370	22.318	52.821	20.873	25.716
9	37.000	21.542	33.775	38.522	38.522	22.191	50.817	20.823	25.557
10	36.108	21.414	33.169	37.844	37.844	22.090	49.214	20.782	25.430

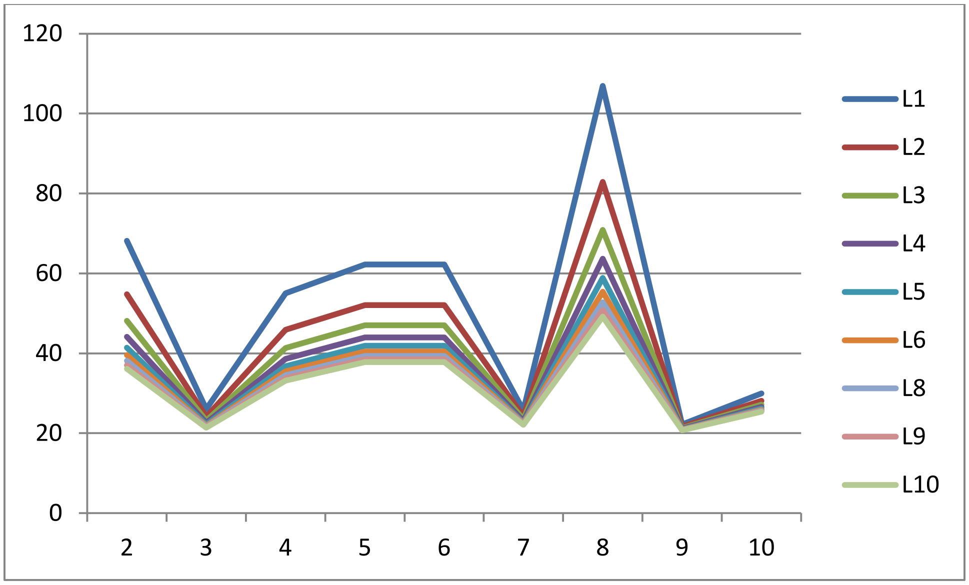


Figure II: Graph of the sensitivity analyses of number of servers against average number of patients

Table II shows the result of average time spent (W) by patients on the system for each day. For example, considering the first column and first row (i.e. two servers for day one), 78.07 indicates that the patient spent 78minutes and 0.079 seconds on the average, both on service and on queue for the first day. The table also depicts the result of when more than two servers are used and their corresponding time spent by patients for each day. While the up and down nature of the sensitivity graph in figure III depicted the change which occurred when there was a change in number of servers against average time spent by patients on the system.

Table II: Results of the sensitivity analyses of time (W) spent by patient on the system for each day

server	W_1	W_2	W_3	W_4	W_5	W_6	W_8	W_9	W_{10}
2	78.079	37.224	65.607	67.675	67.675	36.470	110.81	33.687	40.833
3	62.779	34.482	54.758	56.616	56.616	34.313	85.890	32.771	38.242
4	55.129	33.112	49.334	51.087	51.087	33.235	73.430	32.314	36.946
5	50.539	32.290	46.079	47.770	47.770	32.588	65.954	32.039	36.169
6	47.480	31.741	43.909	45.558	45.558	32.157	60.970	31.856	35.651
7	45.294	31.350	42.359	43.978	43.978	31.849	57.410	31.725	35.281
8	43.655	31.056	41.197	42.794	42.794	31.618	54.740	31.627	35.003
9	42.380	30.828	40.293	41.872	41.872	31.4377	52.663	31.551	34.787
10	41.360	30.645	39.570	41.135	41.135	31.294	51.002	31.489	34.615

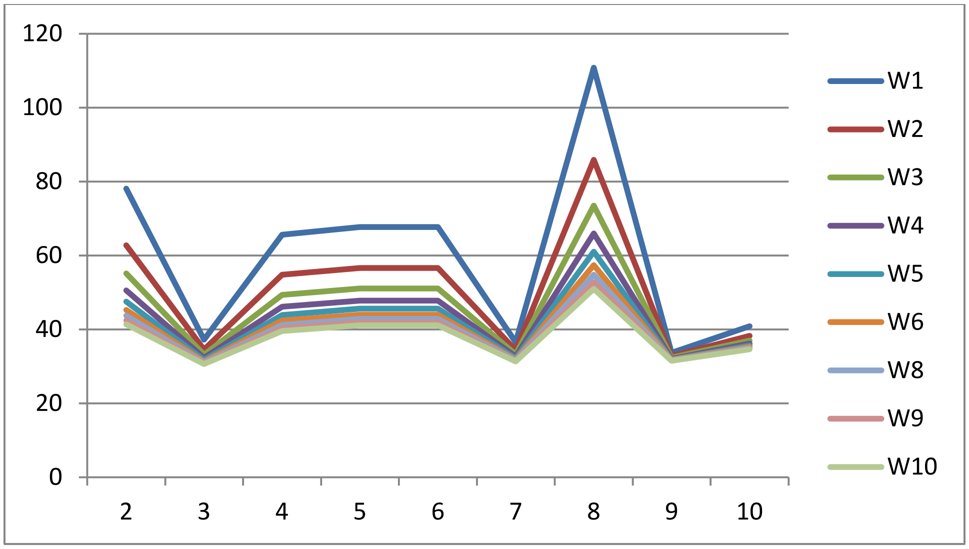


Figure III: Graph of the sensitivity analyses of number of time spent (W) by patient on the system for each day.

Table III: shows the result of average number of patients on the queue (Lq) for each day. For instance, on days one, two, three, four, e.t.c. using two servers, the average number of patients on queue was 40.071, 5.747, 27.282, 30.521, e.t.c. respectively. The table also shows the sensitivity of having more than two servers and the corresponding values of average number of patients on the queue for each of the days. The results are equally demonstrated using figure IV.

Table III; Results of the sensitivities analyses of average number of patients (Lq) on the queue for each day.

Server	Lq_1	Lq_2	Lq_3	Lq_4	Lq_5	Lq_6	Lq_8	Lq_9	Lq_{10}
2	40.071	5.747	27.282	30.521	30.520	4.567	72.1382	1.813	30.521
3	26.714	3.831	18.188	20.347	20.347	3.045	48.0922	1.209	20.347
4	20.036	2.873	13.641	15.260	15.260	2.284	36.0691	0.907	15.260
5	16.028	2.299	10.913	12.208	12.208	1.827	28.8553	0.725	12.208
6	13.357	1.916	9.094	10.174	10.174	1.522	24.0461	0.604	10.174
7	11.449	1.642	7.795	8.720	8.720	1.305	20.6109	0.518	8.720
8	10.018	1.437	6.821	7.630	7.630	1.142	18.0346	0.453	7.630
9	8.9047	1.277	6.063	6.782	6.782	1.015	16.0307	0.403	6.782
10	8.014	1.150	5.457	6.104	6.104	0.913	14.4277	0.363	6.104

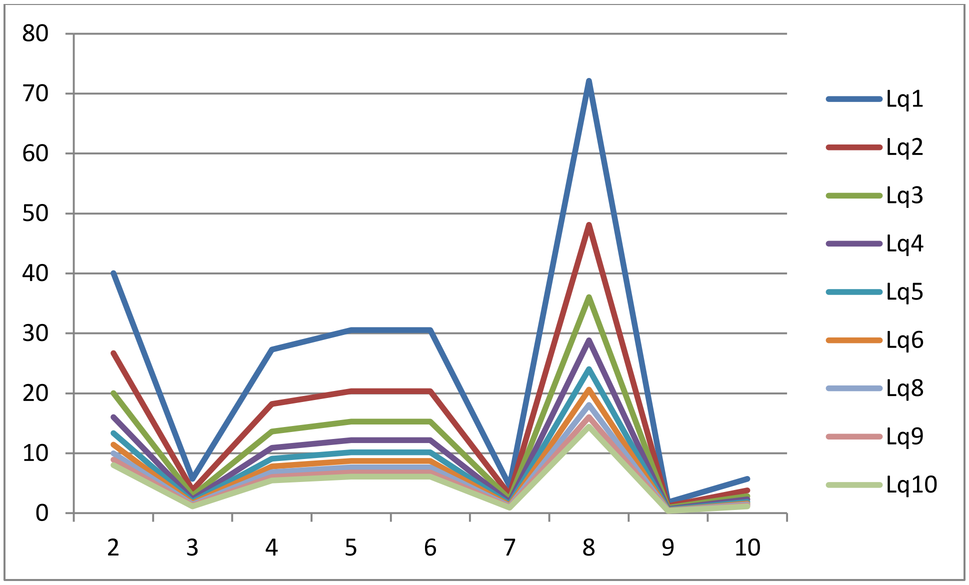


Figure IV: Graph of the sensitivity analyses of average number of patients (Lq) on the queue for each day

Table IV depicts the results of average waiting time of patients on queue before service (Wq) for each day. Using two servers on the first day, the patients spent 45.900 minutes on queue for being attended to by the doctor. With same number of servers on the second, third, fourth to tenth days, 8.224, 32.547, 33.175 minutes e.t.c. was spent respectively by the patients to consult the doctors. Increasing the number of serve as shown in table IV and as depicted graphically in Figure V below, the results show that the time used by patients on the queue reduces compared to when the number of server is decreased.

Table IV: Results of the sensitivity analyses of average waiting time of patient on queue before service (Wq) for each day

server	Wq_1	Wq_2	Wq_3	Wq_4	Wq_5	Wq_6	Wq_8	Wq_9	Wq_{10}
2	45.900	8.224	32.547	33.175	33.175	6.470	74.760	2.747	7.773
3	30.600	5.483	21.698	22.116	22.116	4.313	49.840	1.831	5.182
4	22.950	4.112	16.274	16.587	16.587	3.235	37.380	1.374	3.886
5	18.359	3.290	13.019	13.270	13.270	2.590	29.903	1.099	3.109
6	15.300	2.741	10.849	11.058	11.058	2.157	24.920	0.916	2.591
7	13.114	2.350	9.299	9.478	9.478	1.849	21.360	0.785	2.221
8	11.475	2.056	8.137	8.294	8.294	1.618	18.690	0.687	1.943
9	10.200	1.828	7.233	7.372	7.372	1.438	16.613	0.611	1.727
10	9.180	1.645	6.510	6.635	6.635	1.294	14.952	0.549	1.555

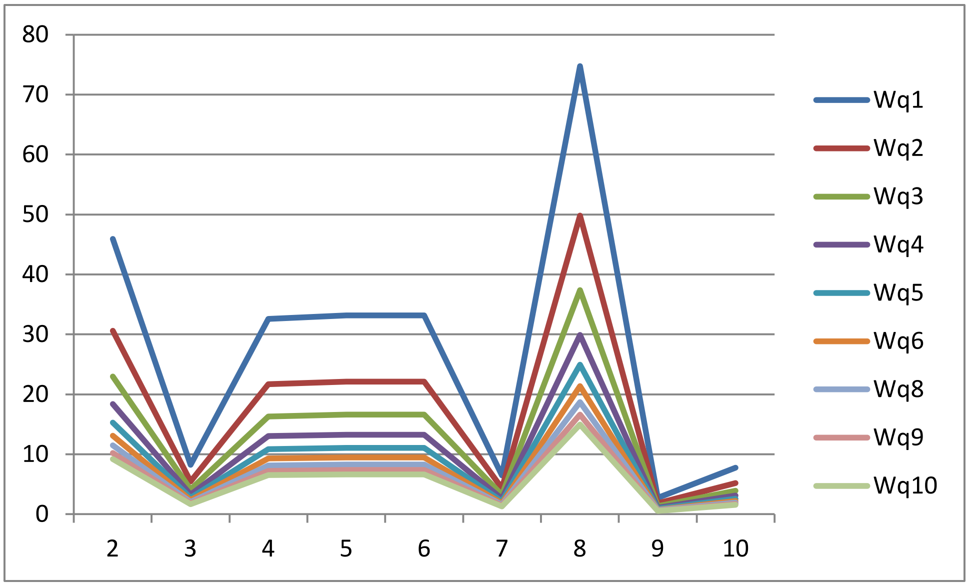


Figure V: Graph of the sensitivity analyses of average waiting time of patient on queuebefore service (Wq) for each day.

It was observed that on day seven, the arrival rate is greater than the product of numbers of servers and service rate which lead to congestion of patients in the system, that is, traffic density is greater than one ($\rho > 1$) which gives no result for day seven.

Therefore, in applying queuing theory, to analyze the observed queue data collected over two weeks period of time from the health service provided to patients at emergency department of Lagos State General Hospital, Odan, Lagos, the results show that the environment of emergency department can move from an orderly one to a complex environment and on to a chaotic environment all in a short period of time and then return to its original state, which means that emergency department did not have an identified peak period. This implies that, two servers are inadequate to serve the numbers of emergency patients during consultation, therefore more hands up to four or six of medical doctors (servers) should be made available in order to reduce the waiting line of patients on queue and throughout the systems, especially on the peak days (when patient are much) of the week of the emergency units.

Summary and Conclusion

When care is provided is as important as what care is provided, satisfaction is seen as a measure of quality; hence enhancing satisfaction is becoming increasingly important. Although, queuing theory, an important tool in decision making can be used in structuring decision making framework and modeling of different healthcare units, general outpatient department, ante-natal clinic, blood banks and other service providing organizations, it is yet to be fully applied and integrated in the emergency department of the hospital. Therefore, the knowledge of queuing model and their effective application during the consultation period of patients by doctors in emergency department of Lagos State General Hospital, Odan, can help improve access to quality health care and reduce loss of lives due to delays. The knowledge of the models can help service management to make decisions that will increase the satisfaction of all concerned parties patients, employees and management.

In conclusion, with the queue model, the emergency department can learn and decide fast on what to do if there is a waiting line and quickly put the situation under control with effective change in number of servers when necessary.

Recommendation

The future of queuing theory in healthcare depends on the commitment and ingenuity of the health care queuing theory community to ensure that improved patient safety, using this tool becomes a reality. Therefore, full integration of queue theory applications into the routine structures and practices should be introduced to reduce the waiting line problems for consulting patient, reduces loss of lives due to delays and improve safety of the health care services in the emergency department of Lagos State General Hospital, Odan. Also, hospital management should ensure that competent and serious minded medical doctors are available for consultation especially during the peak period to reduce the risk during diagnosis of the patient. More also, treatments that are delayed at the beginning of each week due to cleaning and sanitation done in the department could be done earlier before arrival of the patients.

Suggestion for further studies

Despite the contribution of this study to knowledge, there is limitation. The limitation arises from the fact that the study was carried out within two weeks in an emergency department of the hospital due to time frame of the study which could limit the generalization of the findings. Further, since only a unit of the hospital was considered, a queuing model cannot be considered to analyze the patient flow throughout the hospital. Hence, a further study is recommended to cover some selected hospitals over a longer period of time.

References

Babes, M., & Sarma, G. V. (1991). Outpatient queues at the Ibn-Rochd Health Center. The Journal of the Operational Research Society, 42(10), 845-855.

Belson, G. V. (1988), Waiting Times Scheduled Patient in the Presence of Emergency Request, Journal of Operational Management 2(3).

Davis Mark M., Vollmann Thomas E , (1990) "A Framework for Relating Waiting Time and Customer Satisfaction in a Service Operation", Journal of Services Marketing, Vol. 4 Iss: 1, pp.61 – 69

Fomundam S., Jeffrey Herrmann J. (2007), A Survey of Queuing Theory Applications in Healthcare, The Institute for Systems Research, ISRTechnical Report.

Hamdy A. Taha (2007), ‘ Operation Research: An Introduction, 8th Edition, ISBN:0131889230. Published by Pearson Education, Inc., 2007.

Kingman, J. F. C. (2009). "The first Erlang Century—and the Next". Queuing Systems 63: 3–4. doi:10.1007/s11134-009-9147-4.

McLeod RD, Laskowski M, Friesen M.R., Podaima, B.W. (2009), “Agent based models for nosocomial infections: Modeling of hospital-acquired infections within a hospital”. Report for the Public Health Agency of Canada.

Mohammad Shyfur Rahman Chowdhury (2013), ‘Queuing Theory Model Used To Solve The Waiting Line Of A Bank -A Study On Islamic Bank Bangladesh Limited, Chawkbazar Branch, Chittagong’. Asian Journal Of Social Sciences & Humanities. Vol.2, No.3, August 2013.

Nosek, A.R., Wislon, P.J.(2001), Queuing Theory and Customer Satisfaction; A Review of Technology, Trends and Applications To Pharmacy Practice, Hospital Pharmacy, 36 (3); 275-279.

Obamiro J.K.(2010), Queuing Theory and Patient Satisfaction: An Overview of Terminology and Application in Anti -Natal Care Unit, Petroleum-Gas University of Ploiesti Bulletin, Vol. LXII, No. 1/2010,1-11, Economic Sciences Series.

Sanjay K. Bose (2002), ‘Basic Queueing heory’, M/M/-/- Type Queues. https:www.iitg.ernet.in/skbose/qbook/slide_set_3PDF

Sharma S.D. (2010), Operations research: Theory, method and applications, 15thEd. McGraw-Hill Companies, Inc.

Singh V. (2006), Use of Queuing Models in Healthcare, Department of Health Policy and Management, University of Arkansas for medical science.

Taylor, S.(1994), Waiting for Service: The Relationship between Delays and Evaluation of Service, Journal of Marketing, 58(2): 56-69.

Trout, A., Magnusson, A.R. and Hedges, J. R. (2000), “Patient Satisfaction Investigation and the Emergency Department”, Academic Emergency Medicine 7: 695-709.